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import time
import math
from dronekit import connect, VehicleMode, LocationGlobalRelative

connection_string = 'tcp:127.0.0.1:5760'

print("Connecting to vehicle on:", connection_string)
vehicle = connect(connection_string, wait_ready=True)

def arm_and_takeoff(target_altitude):
    print("Performing basic pre-arm checks")
    while not vehicle.is_armable:
        print(" Waiting for vehicle to initialise (GPS/EKF)...")
        time.sleep(1)

    print("Arming motors")
    vehicle.mode = VehicleMode("GUIDED")
    vehicle.armed = True

    while not vehicle.armed:
        print(" Waiting for arming...")
        time.sleep(1)

    print("Taking off!")
    vehicle.simple_takeoff(target_altitude)

    while True:
        current_altitude = vehicle.location.global_relative_frame.alt
        print(" Altitude:", round(current_altitude, 2))
        if current_altitude >= target_altitude * 0.95:
            print("Reached target altitude")
            break
        time.sleep(1)

def get_location_metres(original_location, dNorth, dEast):
    earth_radius = 6378137.0
    dLat = dNorth / earth_radius
    dLon = dEast / (earth_radius * math.cos(math.pi * original_location.lat / 180))
    new_lat = original_location.lat + (dLat * 180 / math.pi)
    new_lon = original_location.lon + (dLon * 180 / math.pi)
    return LocationGlobalRelative(new_lat, new_lon, original_location.alt)

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def get_distance_metres(location1, location2):
    dlat = location2.lat - location1.lat
    dlong = location2.lon - location1.lon
    return math.sqrt((dlat * dlat) + (dlong * dlong)) * 1.113195e5

def goto_waypoint(waypoint, label=""):
    print("Flying to " + label + ": lat=" + str(waypoint.lat) + ", lon=" + str(waypoint.lon))
    vehicle.simple_goto(waypoint)
    while True:
        distance = get_distance_metres(vehicle.location.global_relative_frame, waypoint)
        print(" Distance to " + label + ": " + str(round(distance, 2)) + " m")
        if distance <= 1.0:
            print("Reached " + label)
            break
        time.sleep(1)

TARGET_ALTITUDE = 10
SQUARE_SIDE = 20

arm_and_takeoff(TARGET_ALTITUDE)

home_location = vehicle.location.global_relative_frame

waypoint_1 = get_location_metres(home_location, SQUARE_SIDE, 0)
waypoint_2 = get_location_metres(home_location, SQUARE_SIDE, SQUARE_SIDE)
waypoint_3 = get_location_metres(home_location, 0, SQUARE_SIDE)
waypoint_4 = home_location

print("Starting square flight pattern")
goto_waypoint(waypoint_1, "Waypoint 1 (North)")
goto_waypoint(waypoint_2, "Waypoint 2 (North-East)")
goto_waypoint(waypoint_3, "Waypoint 3 (East)")
goto_waypoint(waypoint_4, "Waypoint 4 (Home)")

print("Square pattern complete. Returning to Launch (RTL)")
vehicle.mode = VehicleMode("RTL")

while vehicle.armed:
    print(" Waiting for landing and disarm...")
    time.sleep(1)

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print("Vehicle has landed and disarmed.")  
vehicle.close()  
print("Mission complete.")
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